

Bias–Variance & Regularization

INFO 521 · student notes · Module 2 · Lecture B · ~7 min read

Lecture A found the U-curve empirically. This lecture explains why it exists (the bias–variance decomposition) and gives a knob to move along it (ridge regression).

Bias and variance

Imagine refitting on many datasets drawn the same way. **Bias** is the systematic miss: a too-simple model is wrong in the same direction on average. **Variance** is the sensitivity: a too-flexible model swings wildly from dataset to dataset, chasing each one's noise. In the repeated-fits picture, a degree-1 model has fits that huddle tightly but miss the truth (high bias, low variance), while a degree-9 model tracks the truth on average but its individual fits fan out wildly (low bias, high variance).

For a fixed input, expected test error splits into three pieces,

$$\mathbb{E} [(y - \hat{f}(\mathbf{x}))^2] = \text{bias}^2 + \text{variance} + \sigma^2,$$

where σ^2 is irreducible noise no model can beat. As complexity rises, bias^2 falls and variance rises; their sum is the U-curve, resting on the σ^2 floor. The best model balances the two — it is not the most flexible one. The full derivation is in the [bias–variance appendix](#).

Ridge regression

Ridge adds a penalty on weight size to the loss,

$$\mathcal{L}_{\text{ridge}}(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \lambda\|\mathbf{w}\|^2,$$

which discourages the large coefficients that drive wiggly overfit curves. It has a closed form that just adds $\lambda\mathbf{I}$ inside the normal equations:

$$\hat{\mathbf{w}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}.$$

That added term lifts every eigenvalue by $\lambda > 0$, making the matrix positive definite and **always invertible** — even with more features than patients, or collinear columns. Ridge hands you, for free, the unique solution Module 1 had to assume.

As λ grows, every coefficient shrinks smoothly toward zero (the coefficient path). The two limits are degenerate: $\lambda \rightarrow 0$ recovers ordinary least squares (the overfit right end of the U), and $\lambda \rightarrow \infty$ crushes the weights to a constant fit (the underfit left end). The useful λ lives in between and is **tuned by cross-validation**, exactly as in Lecture A but with λ as the knob.

The demo

The bias–variance / ridge explorer lets you raise the polynomial degree and watch train and test MSE diverge, toggle **Show 20 fits** to see the variance fan widen at high degree, and slide λ up to pull the fan back in as variance drops and a little bias creeps in.

What to remember

- Test error = bias² + variance + σ^2 ; the U is the sum of the first two.
- Ridge trades a little bias for less variance and is always invertible — uniqueness for free.
- $\lambda \rightarrow 0$ is OLS; $\lambda \rightarrow \infty$ is a constant; CV picks the middle.

Through-lines

Ridge's $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ is the Module 1 uniqueness condition, guaranteed. And the ridge penalty is secretly a **Gaussian prior** on the weights — ridge is MAP estimation, a fact Module 4 makes exact ($\lambda = \sigma^2/\tau^2$).